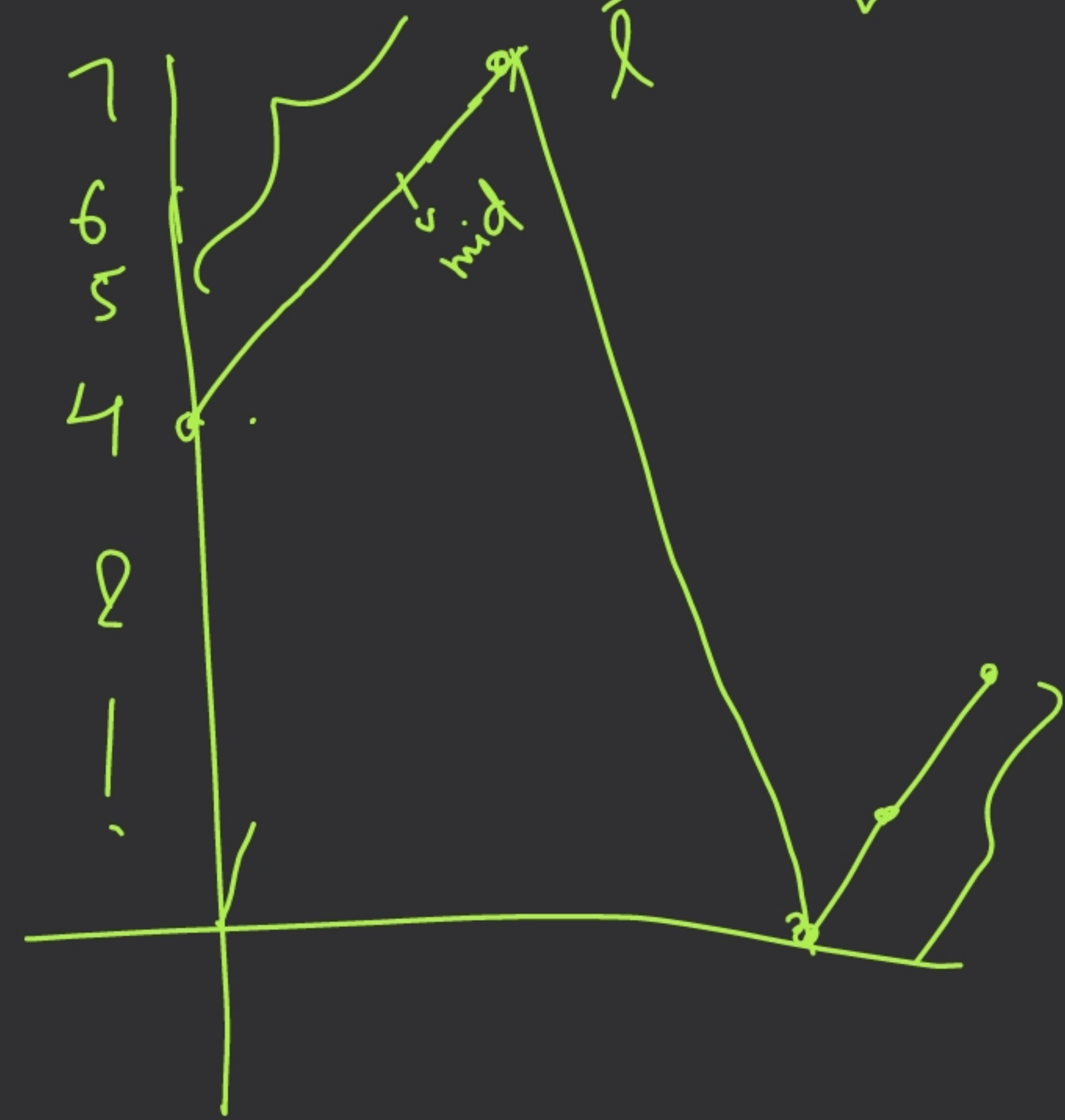


Leetcode 153 :-

mid

ans = INT\_MAX ~~4~~ ~~1~~ 0

[4 5 6 7 0 1 2]



(l  $\rightarrow$  mid) (mid  $\rightarrow$  h)

if arr[l] <= arr[mid] }  $\rightarrow$  (search for left sorted)

otherwise arr[l] > arr[mid]  $\rightarrow$  (right portion)  $\leftarrow$

[0 1 2 3 4 5 6]  
[4 5 6 7 0 1 2]  
8  $\uparrow$  mid

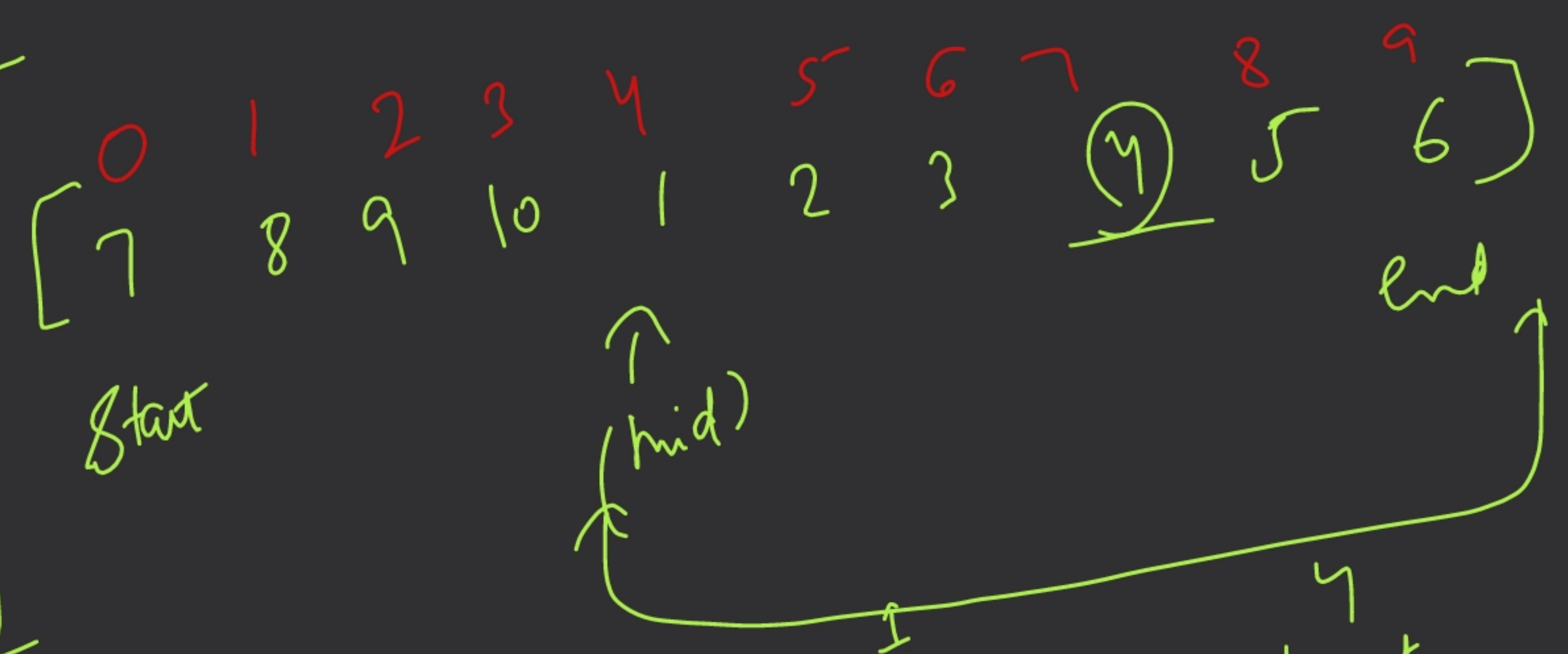
[7 0 1 2]  
 $\uparrow$  8  
 $\uparrow$  m  $\uparrow$  e

0  
 $\uparrow$   
5  
 $\uparrow$   
2  
 $\uparrow$   
m

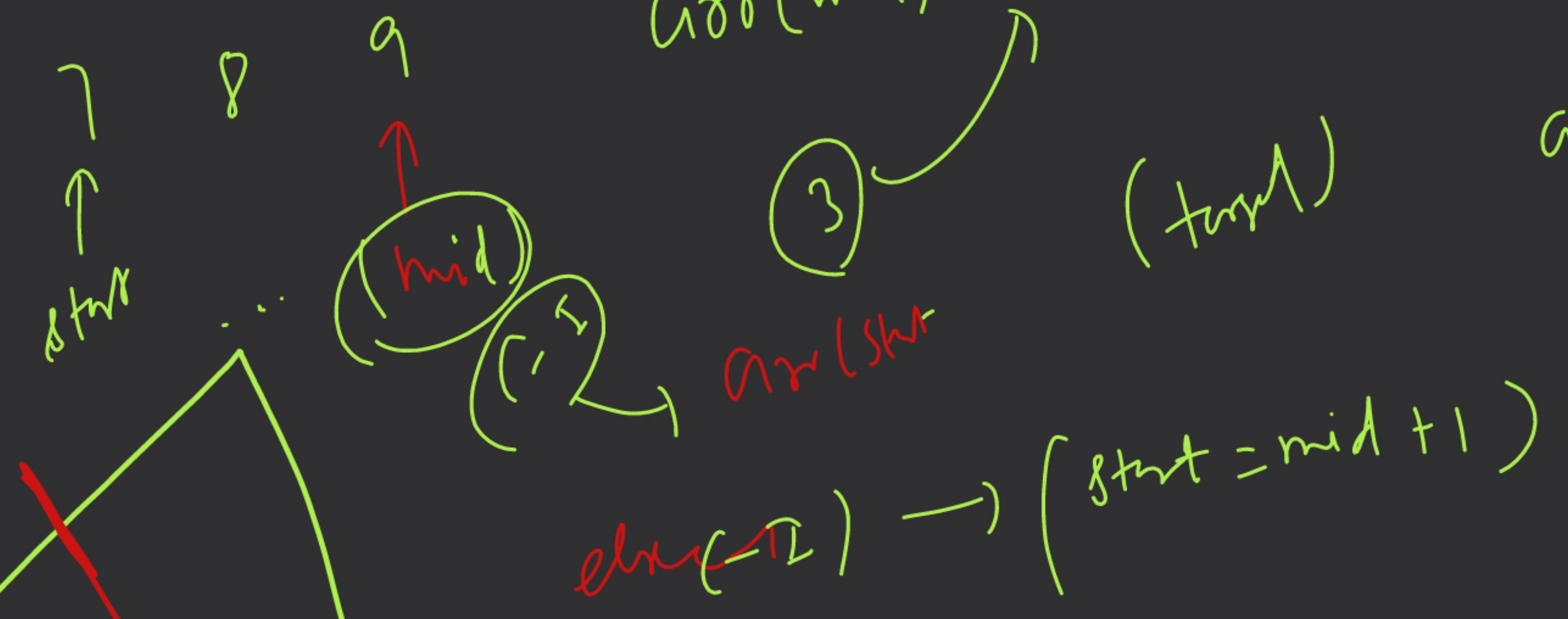




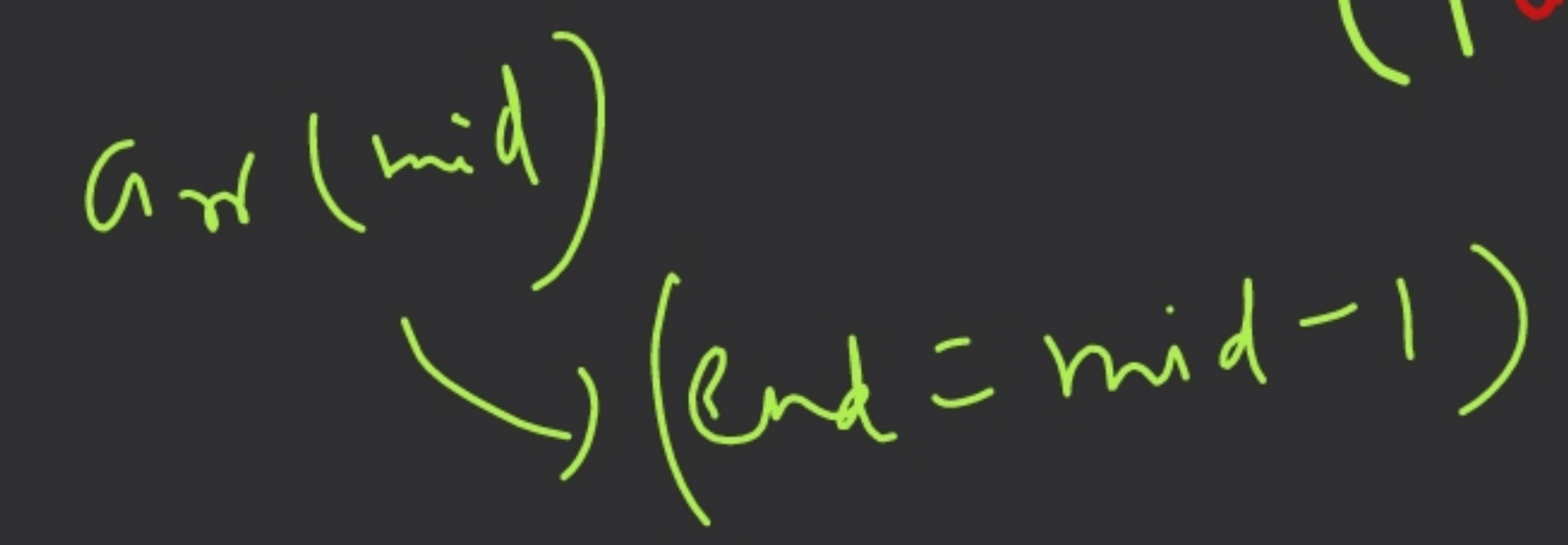
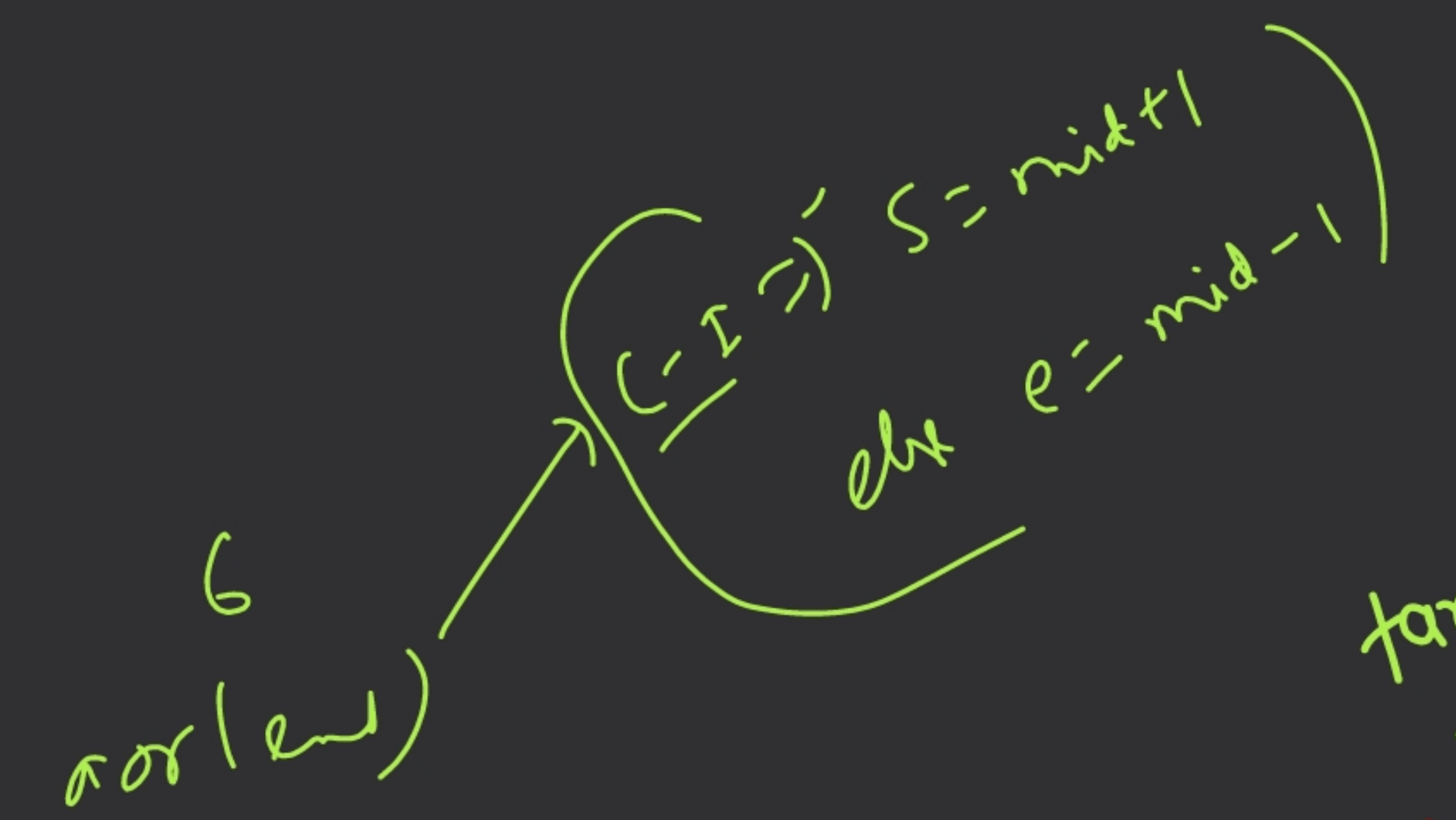
Leet code 33 :-



(target = 8)



not possible (eliminated)



end = mid - 1

(arr[mid] = 10)

target in b/w

(target = 8)

(eliminating)

target outside (target > n)

(start = mid + 1)





```
f(arr, n, target) {  
    int start = 0, end = n - 1;
```

```
    while (start <= end) {
```

```
        mid =  $\frac{start + end}{2}$ ;
```

```
        if (arr[start] <= arr[mid]) {
```

```
            if (arr[start] <= target && target <= arr[mid) {
```

```
                end = mid - 1;
```

```
            } else start = mid + 1;
```

```
        }
```

// right

```
    } else { if (arr[mid] <= target &&  
        target <= arr[end) {
```

```
        start = mid + 1;
```

```
    } else end = mid - 1;
```

```
    }
```

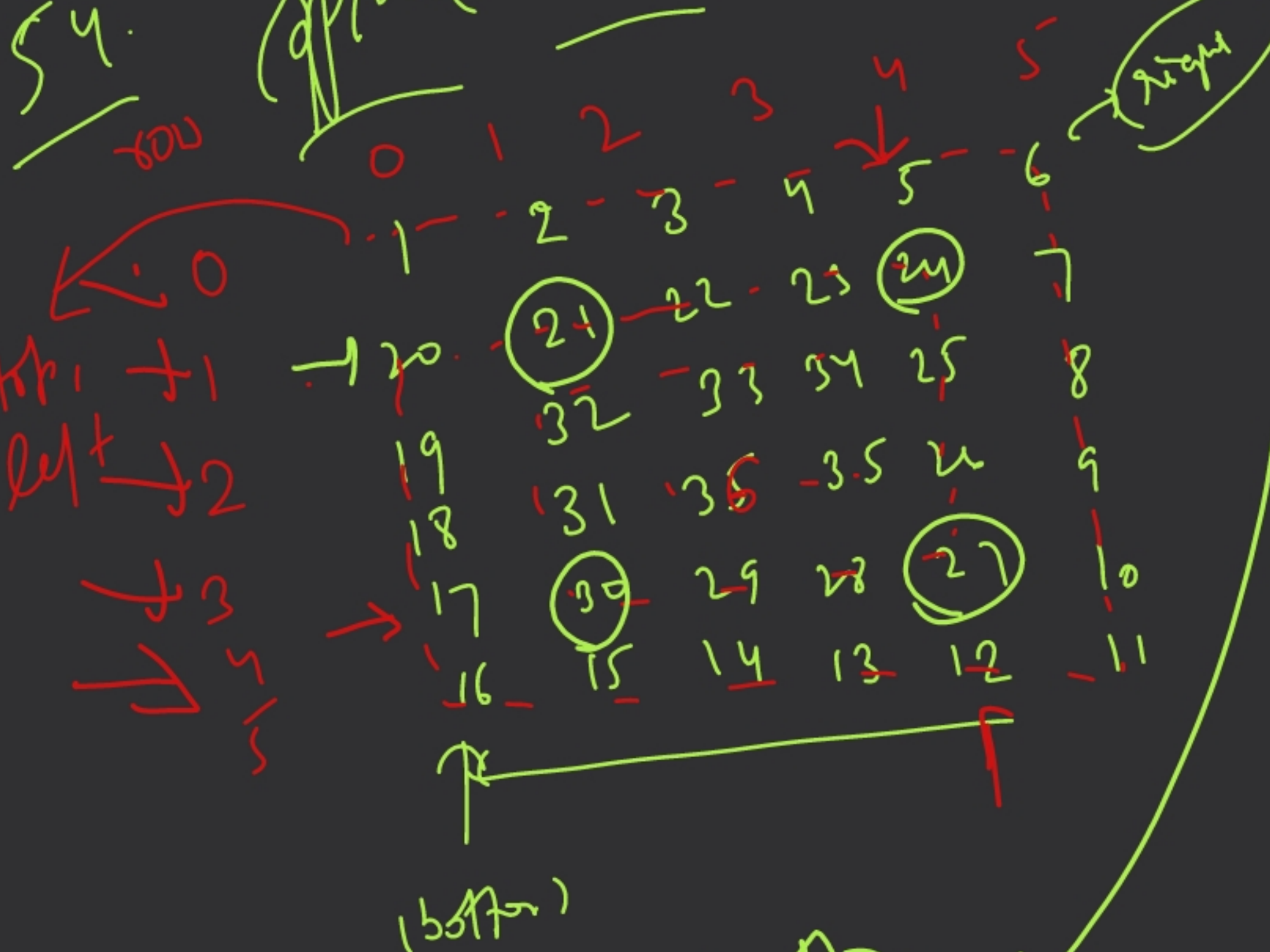
```
}
```

```
return -1;
```

```
}
```

left  
portion  
→

54. Spiral Matrix:-



$top = 0, left = 0, bottom = n - 1, right = m - 1$   
 $list < Integer > spiral = new ArrayList<>();$

```

for (int i = left; i <= right; i++) {
    spiral.add(arr[top][i]);
}
top++;

for (int i = top; i <= bottom; i++) {
    spiral.add(arr[i][right]);
}
right--;

for (int i = right; i >= left; i--) {
    spiral.add(arr[bottom][i]);
}
bottom--;

```

if (top <= bottom)

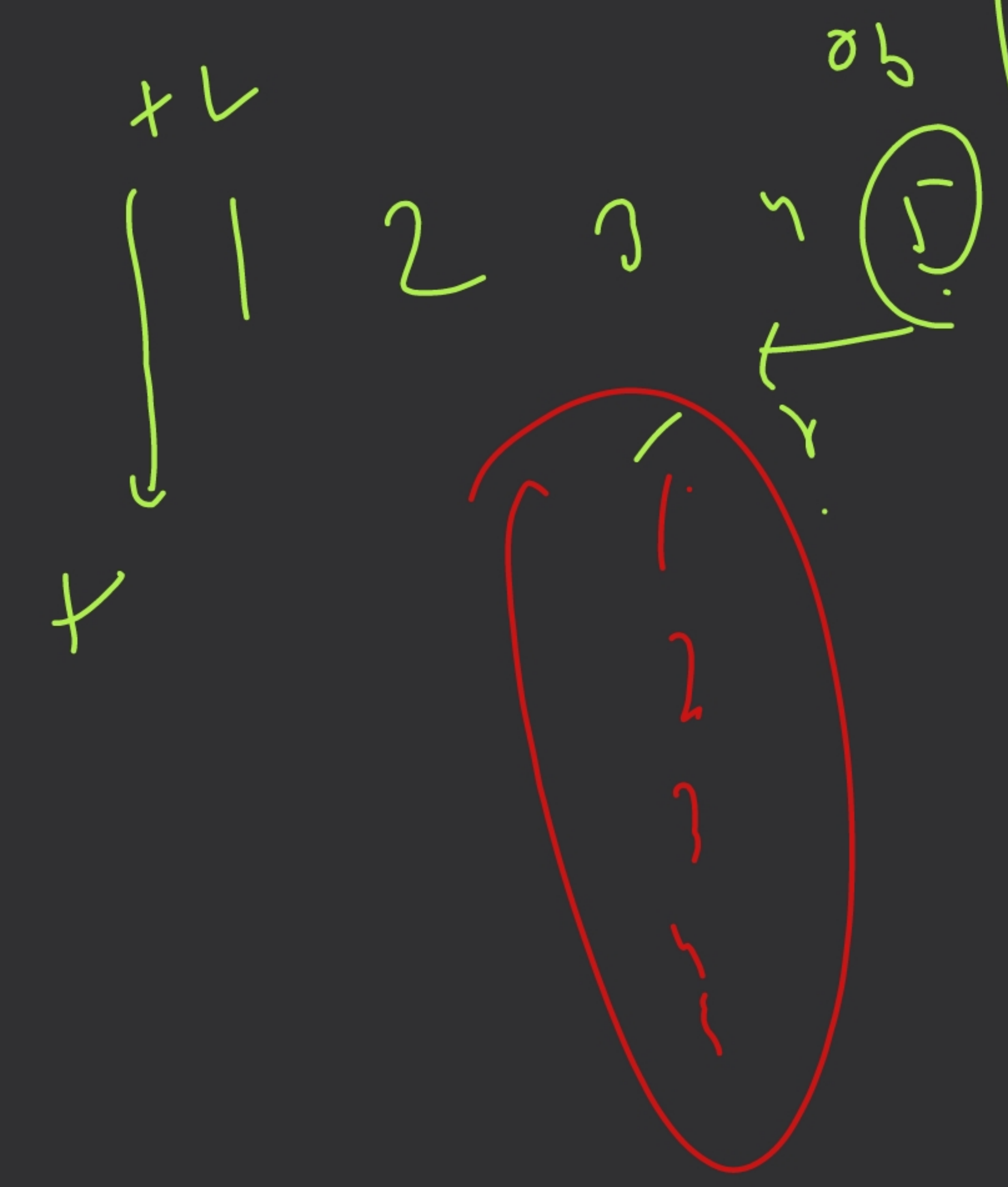
```

if (left <= right) {
    for (int i = bottom; i >= top; i--) {
        spiral.add(arr[i][left]);
    }
    left++;
}

```

List < Integer > spiral =  
 new ArrayList<>();

while (top <= bottom  
 && left <= right) {



HW.

